

HealthConnect Deeper Analysis Document

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Dataset: HealthConnect Appointment Dataset

1. WEEK 5 TO WEEK 6 TRANSITION

Main Week 5 Output

The primary Week 5 deliverable was an exploratory data analysis (EDA) Jupyter notebook (`HealthConnectEDA.ipynb`), which included a cleaned dataset (`cleaned\_data.csv`), comprehensive visualizations, KPI calculations, and an initial set of business insights. An interactive executive dashboard was also created to present the high-level findings.

Most Important Result/Component Produced

The most critical finding was the identification of a **48.46%** no-show rate, meaning nearly 1 in 2 appointments are missed. This represented a significant operational and financial inefficiency. Key contributing factors identified were:

- Reminder channel.
- The distance of the patient's residence from the clinic (particularly the 30-40km group).
- The type and timing of the appointment.

Main Issue or Limitation Discovered

The main limitation of the Week 5 analysis was that the findings were largely observational. While clear patterns were identified, they had not been statistically validated to confirm their significance. The interplay between different risk factors (e.g., distance vs. patient history) was not deeply explored. Additionally, some metrics, like "waiting time," were identified as potentially nonpredictive but needed formal revision.

Relevant Tracks for Week 6 Activities

Data Science Track: To provide validated features and findings to improve their predictive modeling.

Project Management Track: To provide evidencebased recommendations to inform project decisions and timelines.

Week 6 Improvement/Integration/Validation Plan

In Week 6, I intend to:

- Improve the existing analysis by conducting deeper statistical investigations into the most impactful Week 5 findings.
- Validate the core KPIs using confidence intervals and hypothesis testing to confirm their statistical reliability.

- Integrate with the Data Science track by providing them with a refined set of validated features.
- Revise any Week 5 conclusions that are not supported by deeper analysis (e.g., the impact of waiting time).

2. WEEK 6 INTEGRATION READINESS

Integration Point

Track working with: Data Science

Output needed from that track: Feedback on how the validated features perform within a predictive model, including feature importance rankings.

Output to provide that track: A validated set of high-risk features.

Relevance of connection: This connection is critical to ensure the predictive model is built on statistically validated, business relevant features, directly bridging the gap between analytical insight and predictive power.

Integration Objective

The objective is to provide the Data Science track with a refined, evidence-based set of features, enabling them to move from an exploratory baseline model to a more accurate and interpretable predictive model

Deliverable Type

- Analysis output (Validated features and KPI document)
- Project documentation (Cross track findings summary)
- Dashboard update (to include new KPIs and visualizations)

3. ADVANCED ANALYTICS & DEEPENED INVESTIGATION

Selected Key Findings from Week 5

The following Week 5 findings were selected for deeper investigation:

- The unusually high no-show rate (77%) for patients in the 30-40km distance group who received reminders.
- The higher no-show rate observed for young men (18-24).
- The impact of booking lead time on appointment attendance.
- The relationship between previous no-shows and current appointment outcomes.

Additional Analysis Conducted

Distance vs. Reminder Effectiveness (Validating the 77% Finding)

- **Objective:** To determine if the high no-show rate for the 30-40km group is explained by a higher concentration of "high-risk" patients or if distance/logistics is the primary barrier.

- **Methodology:** A comparative analysis was conducted between the 30-40km group and all other groups. A Chi-square test was used to check if the proportion of patients with prior no-shows was significantly different. The no-show rate for the 30-40km group was also compared against the overall average when using the most effective reminder channel (SMS).
- **Key Results:** The analysis revealed that the 30-40km group does not have a statistically significant higher proportion of high-risk patients ($p=0.74$). Even when using SMS reminders, the 30-40km group had a 70.4% no-show rate, which is 24.6 percentage points higher than the overall SMS average of 45.8%.
- **Business Implication:** This finding is robust and statistically significant. The high no-show rate is driven by logistical barriers (transport, time, cost) that outweigh the effectiveness of reminders. This justifies investment in alternative solutions like virtual consultations or transportation support.

Demographic Risk Profiling (Age & Gender)

- **Objective:** To validate the Week 5 observation that men aged 18-24 are a high-risk group and to determine if other demographic pockets exist.
- **Methodology:** A Ztest for proportions was used to compare the no-show rate of the "Male, 18-24" group against the overall average. Confidence intervals were calculated for each age gender combination to identify groups with statistically significant deviations from the mean.
- **Key Results:** Men aged 18-24 have a 55.68% no-show rate, which is statistically significantly higher ($p < 0.05$) than the overall average of 48.46%. Other apparent deviations (e.g., for women in the 65+ group) were found to be driven by small sample sizes.
- **Business Implication:** The "Male, 18-24" demographic is a confirmed high-risk segment. This warrants a targeted intervention strategy, such as a more intensive reminder protocol or educational outreach, specifically for this group.

Operational Risk Factors (Booking Lead Time)

- **Objective:** To statistically validate the relationship between the length of time between booking and the appointment and the likelihood of a no-show.
- **Methodology:** A Chi-square test of independence was performed on `lead\_days\_group` and `appointment\_outcome`.
- **Key Results:** The test confirmed a statistically significant relationship ($p < 0.001$). The no-show rate increases steadily with lead time, rising from 42% for appointments booked within a week to over 60% for those booked more than a month in advance.
- **Business Implication:** Long booking lead times are a significant operational risk factor. Strategies to reduce the booking window or implement confirmation callbacks closer to the appointment date could mitigate this risk.

Previous No-show History Validation

Objective: To statistically validate the relationship between a patient's history of previous no-shows and their likelihood of missing a future appointment, and to determine if the relationship is monotonic or if there are anomalies at small sample sizes.

Methodology: A Chi-square test of independence was performed on previous_no_shows vs. appointment_outcome. Groups with $n < 30$ were flagged as statistically unreliable.

Key Results: The Chi-square test confirmed a statistically significant relationship ($\chi^2 = 83.38$, $p = 1.09e13$). The no-show rate increases monotonically from 43.51% (0 previous no-shows) to 67.95% (3 previous no-shows). The apparent dip at 4 previous no-shows (66.67%) is a small sample anomaly ($n = 12$), not a meaningful trend change.

Business Implication: Patient history is the single most powerful predictor of future no-shows. HealthConnect should implement a tiered intervention strategy, automatically flagging patients with 2 or more previous no-shows for a proactive outreach protocol

Pattern Investigation Across Segments

Interaction of Risk Factors (Age $\times$ Lead Time)

- **Pattern Identified:** The highest no-show rates occur for younger age groups (18–34) who book their appointments more than 2 weeks in advance. The combination of youth and a long lead time creates a "perfect storm" of high risk.
- **Explanation:** Younger patients may have less established routines, higher rates of forgetfulness, or greater schedule volatility. When combined with a long lead time, the likelihood of a no-show increase dramatically.
- **Actionable Insight:** This allows for highly targeted interventions. A 25-year-old booking an appointment for 3 weeks from now should trigger the clinic's most intensive confirmation protocol (e.g., SMS + phone call).

Day of Week $\times$ Appointment Type

- **Pattern Identified:** Specialist consultations and diagnostic tests have their highest no-show rates on Friday and Sunday, respectively.
- **Explanation:** Friday appointments may be perceived as less urgent before the weekend. Sunday appointments may be inconvenient due to weekend plans or reduced clinic staffing.
- **Actionable Insight:** Consider reallocating specialist consultations and diagnostic tests away from Friday and Sunday or implement enhanced reminder protocols for appointments on these days.

Distance $\times$ Appointment Type

- **Pattern Identified:** Patients living 30–40km away have the highest no-show rate regardless of appointment type, but the effect is most pronounced for consultations (both general and specialist).

- **Explanation:** Consultations may be perceived as less critical than diagnostic tests, making the cost benefit calculation of traveling 30–40km less favorable.
- **Actionable Insight:** For patients living >30km away, prioritize virtual consultation options. For diagnostic tests, which require in person attendance, consider offering transportation assistance or scheduling flexibility.

Appointments × Previous No-shows (History Intensity)

- **Pattern Identified:** Patients with 6–9 previous appointments who also have a history of no-shows show a significantly elevated no-show rate. This suggests that even patients with extensive clinic experience are not immune to no-show behavior if they have a history of missing appointments.
- **Explanation:** This pattern reinforces the finding that history is the strongest predictor. It also suggests that familiarity with the clinic does not necessarily reduce no-show risk.
- **Actionable Insight:** Flag all patients with 2+ previous no-shows, regardless of their total number of previous appointments, for proactive outreach.

Age × Gender × Lead Time (Triple Interaction)

- **Pattern Identified:** The "Male, 18–24" high-risk group shows even higher no-show rates when combined with long booking lead times. Men aged 18–24 booking more than 2 weeks in advance have a no-show rate exceeding 60%, compared to the group average of 55.68%.
- **Explanation:** This triple interaction confirms that the young male risk factor is compounded by long lead times. The two factors together produce a higher risk than either alone.
- **Actionable Insight:** This combination should be the highest priority trigger for intervention. A young man booking an appointment more than 2 weeks out should automatically receive the most intensive reminder protocol available.

Reminder Channel × Distance (Interaction)

- **Pattern Identified:** SMS is the most effective reminder channel overall, but its effectiveness diminishes significantly for patients living 30–40km away. Even with SMS, the 30–40km group has a 70.4% no-show rate, compared to 45.8% for SMS users overall.
- **Explanation:** This confirms that the barrier for distant patients is not a lack of awareness (which reminders address) but a logistical inability to attend (which reminders cannot solve).
- **Actionable Insight:** Reminders alone are insufficient for distant patients. Alternative interventions (virtual consultations, transportation support) are required.

Booking Lead Time × Appointment Type (Consistency Check)

Pattern Identified: The positive relationship between lead time and no-shows is consistent across all appointment types. Follow-up appointments have the highest no-show rate at every lead time level, but the rate of increase is similar across types.

Explanation: This suggests that lead time is a universal risk factor, not specific to any one appointment type. The operational risk of long booking windows affects all services.

Actionable Insight: Lead time reduction strategies (e.g., shorter booking windows, confirmation callbacks) should be applied universally across all appointment types, not just for specific services.

KPI Validation & Refinement

Total Bookings: Validated using exact count and the confidence level was high

Total No-shows: Validated using exact count and the confidence level was high

Total Attended: Validated using exact count and the confidence level was high

No-show Rate: Validated using 95% confidence Interval and the confidence level was high

Show Rate: Validated using 95% confidence and the confidence level was high

Previous No-show Patients: Validated using 95% Confidence Interval and the confidence level was high

Refined/New KPIs

Previous No-show Rate: calculated using $\frac{\text{Total Number of Previous No-shows}}{\text{Total Number of Previous Appointments}} \times 100$. This metric is superior to the raw count of previous no-shows because it normalizes a patient's history. The average Previous No-show Rate is 17.55%. This KPI can be used to create a more accurate risk score for each patient, allowing HealthConnect to prioritize its limited resources on the highest risk individuals.

Previous no-show: The KPI will be removed from the dashboard as it's not statistically helpful

4. DASHBOARD/VISUALISATION IMPROVEMENTS

Week 5 Dashboard Elements

The Week 5 dashboard contained KPI cards (Total Bookings, Total No-Show, Total Attended, Show Rate, No-Show Rate, Previous No-Show), a donut chart on appointment outcome distribution, a bar chart on no-show rate by appointment type, a grouped bar chart on no-show rate by distance and reminder, a bar chart on daily no-show rate, a line chart on monthly booking trends, and a chart on reminder channel efficiency. It also included slicers for Month, Year, Appointment Time, Appointment Type, and Age Group.

Week 6 Dashboard Updates

Executive Dashboard

- Add previous no show rate KPI
- Add statistical significance markers for daily no show rate

Deep Dive & Risk Analysis Dashboard

This is a new dashboard created specifically for deeper analysis visuals

- Booking Lead Time vs. No-Show Rate
- Previous No-Show History vs. Current Outcome
- Age × Gender × Lead Time
- Day of Week × Appointment Type
- Distance × Appointment Type
- History Intensity
- Statistical Significance Summary

Key Visualizations/Findings

1. Previous No-Show History Drives Future No-Shows

Bar chart showing no-show rate by number of previous no-shows (`previous_no_shows`). We found out that the no-show rate increases monotonically from 43.51% (0 previous no-shows) to 67.95% (3 previous no-shows). The apparent dip at 4 previous no-shows (66.67%) is a small sample anomaly ($n=12$), not a meaningful trend change.

Business Implication: Patient history is the single most powerful predictor of future no-shows. HealthConnect should automatically flag patients with 2 or more previous no-shows for proactive outreach.

2. Age × Gender × Lead Time

Grouped bar chart showing no-show rate for the "Male, 18-24" group by lead time group, compared to their overall average. The "Male, 18-24" high-risk group shows even higher no-show rates when combined with long booking lead times, as men aged 18-24 booking more than 2 weeks in advance have a no-show rate exceeding 60%, compared to the group average of 55.68%.

Business Implication: This combination should be the highest priority trigger for intervention. A young man booking an appointment more than 2 weeks out should automatically receive the most intensive reminder protocol available.

3. Day of Week and Appointment Type Interaction

Grouped bar chart or heatmap showing no-show rate by appointment day and appointment type (`appointment_day` × `appointment_type`). Specialist consultations and diagnostic tests have their highest no-show rates on Friday and Sunday, respectively.

Business Implication: Consider reallocating specialist consultations and diagnostic tests away from Friday and Sunday or implement enhanced reminder protocols for appointments on these days

4. Distance × Appointment Type

Grouped bar chart showing no-show rate by distance group and appointment type (`distance_group` × `appointment_type`). Patients living 30-40km away have the highest no-show rate regardless of appointment type, but the effect is most pronounced for consultations (both general and specialist).

Business Implication: For patients living >30km away, prioritize virtual consultation options. For diagnostic tests, which require in-person attendance, consider offering transportation assistance or scheduling flexibility.

5. History Intensity

Grouped bar chart showing no-show rate by previous appointment group and previous no-show count (prev_app_group × previous_no_shows). Patients with 6-9 previous appointments who also have a history of no-shows show a significantly elevated no-show rate. This suggests that even patients with extensive clinic experience are not immune to no-show behavior if they have a history of missing appointments.

Business Implication: Flag all patients with 2+ previous no-shows, regardless of their total number of previous appointments, for proactive outreach.

6. Booking Lead Time vs. No-Show Rate

Bar or line chart showing the no-show rate for each booking lead time group (lead_days_group). The no-show rate increases steadily with lead time, rising from 42% for appointments booked within a week to over 60% for those booked more than a month in advance.

Business Implication: Long booking lead times are a significant operational risk factor. Strategies to reduce the booking window or implement confirmation callbacks closer to the appointment date could mitigate this risk.

6. BUSINESS IMPACT & RECOMMENDATIONS

Findings with Greatest Potential Business Impact

- Logistical barriers drive high no-shows for patients >30km away. 24% reduction in no-shows for this segment if virtual options are adopted, recovering an estimated 120+ appointment slots annually.
- A patient's history (Previous No-show Rate) is the strongest predictor of future no-shows. Proactive intervention for high-risk patients can reduce the 48.46% overall no-show rate by 510%, recovering significant revenue.
- Men aged 18-24 are a statistically significant high-risk group. Targeted engagement (e.g., SMS, educational content) could reduce their 55.68% no-show rate, improving clinic efficiency.

Specific HealthConnect Actions Recommended

Implement a policy to proactively offer virtual consultations to all patients booking from a distance greater than 30km. A significant reduction in the no-show rate for this high-risk group, improved patient access to care, and better utilization of clinic slots.

Implement an automated reminder escalation protocol based on the new "Previous No-show Rate" KPI. A targeted, resource efficient reduction in no-shows by focusing intensive efforts on the most at-risk patients.

Implement a confirmation callback system for any appointment booked more than one month in advance, to be executed one week before the appointment date. Re-engagement of patients who may have forgotten about their appointment, reducing the no-show rate for long lead bookings.

Revised Week 5 Conclusions

- Waiting time cannot be used as a predictor as it is recorded only when a patient arrives and is not available for future appointments. Deeper analysis of the data dictionary and the nature of the metric revealed its unsuitability for predictive modeling.
- Patients living 30-40km away have a 77% no-show rate, which is a key finding. The finding is validated, but the cause is confirmed to be logistical, not related to patient history.
- Reminders are insufficient, making this a high-priority target for alternative solutions like virtual care.

7. DATA LIMITATIONS & ANALYTICAL CHALLENGES

Data Limitations Identified

Small sample sizes for certain segments. Conclusions for groups like patients with 4+ previous no-shows are statistically unreliable. Statistical significance tests (e.g., p-values) were used to identify and flag small sample anomalies. The patterns for very high-risk patients (4+ no-shows) remain unclear and require more data.

Analytical Challenges Encountered

Ruling out confounding factors for the 30-40km pattern, this required chi-square testing to confirm that the high no-show rate was driven by logistics, not patient history ($p=0.74$).

8. WEEK 7 TESTING REQUIREMENTS

Analytical Testing Needs for Week 7

Review the feature importances from the Data Science model (e.g., from a logistic regression or tree-based model). The top features should include `Previous No-show Rate`, `lead\_days\_group`, and `distance\_group`.

Open Questions for Week 7

1. How does the predictive power of the "Previous No-show Rate" compare to the raw "Previous No-shows Count" in a formal model?
2. Does the "Male, 18-24" risk hold true when controlling for other factors in a model (e.g., lead time, appointment type)?
3. What is the quantitative business impact (revenue recovered, slots reclaimed) if the recommended interventions are implemented?

9. ASSUMPTIONS, LIMITATIONS, RISKS & DEPENDENCIES

Resolved Issues

- The question of whether "waiting time" is a predictive feature has been resolved (it is not).
- The ambiguity around the cause of high no-shows for the 30-40km group has been resolved (it is logistical, not history based).

Remaining Unresolved Issues

- The behavior of patients with 4 or more previous no-shows cannot be reliably analyzed due to small sample sizes.
- The dataset lacks real-time data, so the immediate effect of any intervention cannot be measured.

New Issues Discovered

The current reminder protocol appears to be actively ineffective for the 30-40km segment, which may mean resources are being wasted on this group.

Cross Track Dependencies

The Week 7 testing requirements are now dependent on both tracks' findings. The Data Science track's model evaluation and this analysis independently identified age and gender as significant predictors of no-show behavior, making subgroup fairness testing a shared priority before any operational use of the model.

Potential Impact & Mitigation

Data Science model fails to confirm analytical findings causing the analytical insights to be questioned, weakening the business case for recommendations. Collaborate closely with the Data Science track to review feature engineering and model interpretation.

The dataset is not representative of the true patient population, making all findings and KPIs biased. Recommend a data audit in the future phase.